Text Large Model + Robotic Arm Control

Function Description

After the program runs, after waking up the voice module, speak a series of robotic arm control commands. The large model will plan the corresponding action functions for the action commands and execute all instructions in sequence.

Startup

Open the terminal and enter the following command:

```
ros2 launch largemodel largemodel_control.launch.py
```

Currently, the following types of robotic arm control commands are available:

- Control individual servo: Set servo x to y degrees, x ranges from 1-6, y ranges from 0-180. For example, set servo 1 to 120 degrees.
- Control individual robotic arm actions: Robotic arm up/down/left/right, grip posture, tracking posture. For example, set the robotic arm to grip posture, extend the robotic arm upward;
- Control robotic arm motion groups: Robotic arm dance/clap/nod/shake head. For example, please nod;
- Change robotic arm end effector position: Adjust robotic arm up/down/left/right/forward/backward by x centimeters. For example, adjust the robotic arm upward by 3 centimeters

After waking up the module, give a robotic arm control command, reference the following example:

```
First, adjust the robotic arm upward by 3 centimeters, wait 3 seconds, then extend the
robotic arm downward, stay for 3 seconds, then have the robotic arm shake its head, pause
for 3 seconds, and finally set servo 1 to 150 degrees.
```

```
[asr-5] [INFO] [1775200637.708199902] [asr]: -
[asr-5] [INFO] [1775200637.768089953] [asr]: -
[asr-5] [INFO] [1775200637.828238155] [asr]: -
[asr-5] [INFO] [1775200637.888032835] [asr]: -
[asr-5] [INFO] [1775200641.939551637] [asr]: XunFei ASR Finish.
[asr-5] [INFO] [1775200641.940254604] [asr]:  First adjust the robotic arm upward by three centimeters, wait three seconds, then extend the robotic arm down
ward stay for three seconds, then have the robotic arm, shake its head, pause for three seconds, and finally set, serve one to 150 degrees.
[asr-5] [INFO] [1775200641.941318332] [asr]: ☺okay, let me think for a moment...
[model_service-3] [INFO] [1775200658.304371249] [model_service]: Decision making AI planning:Perform the following steps in sequence:
[model_service-3]
[model_service-3] 1. Call the function to move the robotic arm upward by three centimeters.
[model_service-3] 2. Call the function to wait for three seconds.
[model_service-3] 3. Call the function to move the robotic arm downward by three centimeters.
[model_service-3] 4. Call the function to wait for three seconds.
[model_service-3] 5. Call the function for the robotic arm to shake its head.
[model_service-3] 6. Call the function to wait for three seconds.
[model_service-3] 7. Call the function to adjust joint 6 to 150 degrees.
[model_service-3] [INFO] [1775200660.384421280] [model_service]: "json_str": {"response": "Got it! I'll follow your instructions step by step, like a well-c
horeographed dance. Let's begin!", "action": ["arm_move('up', 3)"]}
[model_service-3] [INFO] [1775200660.385155081] [model_service]: "action": ["arm_move('up', 3)"], "response": Got it! I'll follow your instructions step by
step, like a well-choreographed dance. Let's begin!
[action_service-4] [INFO] [1775200660.392162635] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200660.392831194] [action_service]: "actions": ["arm_move('up', 3)"]
[action_service-4] [INFO] [1775200668.851051255] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.855143961] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.859182242] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.865376218] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.869467739] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.873896466] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200668.874478728] [action_service]: Dir: up
[action_service-4] [INFO] [1775200668.875134343] [action_service]: Dist: 3
[action_service-4] [INFO] [1775200668.876008662] [action_service]: cur_joints: [0, 0, 0, 0, 0, 0]
[kinemarics_dofbot-1] [WARN] [1775200668.938218170] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[kinemarics_dofbot-1] [INFO] [1775200668.939471602] [rclcpp]: fk sent***
[action_service-4] [INFO] [1775200670.878584568] [action_service]: CurEndPos: [0.0020449992274857477, -0.06338999792660725, 0.040402443637541426, 1.57080366
10372729, 3.650393812038131e-08, -1.5707926657577578]
[action_service-4] [INFO] [1775200670.879199830] [action_service]: request: dofbot_interface.srv.Kinemarics_Request(tar_x=0.0020449992274857477, tar_y=-0.06
338999792660725, tar_z=0.07040244363754142, roll=1.5708036610372729, pitch=3.650393812038131e-08, yaw=-1.538546913711967, cur_joint1=0.0, cur_joint2=0.0, cu
r_joint3=0.0, cur_joint4=0.0, cur_joint5=0.0, cur_joint6=0.0, kin_name='ik')
[kinemarics_dofbot-1] [WARN] [1775200670.882166179] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[action_service-4] [INFO] [1775200672.387951878] [action_service]: joints: [0.0, 163, 195, 182, 90, 30]
[action_service-4] [INFO] [1775200674.389933632] [action_service]: Moving.
[action_service-4] [INFO] [1775200674.390737692] [action_service]: self.combination_mode: False
[action_service-4] [INFO] [1775200674.391390936] [action_service]: self.interrupt_flag: False
[action_service-4] [INFO] [1775200674.392014143] [action_service]: Move done.
[action_service-4] [INFO] [1775200674.392819462] [action_service]: Published message: Robot feedback: Execution arm_move_done() completed
[model_service-3] [INFO] [1775200676.547303377] [model_service]: "json_str": {"response": "Alright, I've moved upward by 3 centimeters. Now, let's take a li
ttle pause—just like taking a breath before the next move.", "action": ["wait(3)"]}
[model_service-3] [INFO] [1775200676.547890991] [model_service]: "action": ['wait(3)'], "response": Alright, I've moved upward by 3 centimeters. Now, let's
take a little pause—just like taking a breath before the next move.
[action_service-4] [INFO] [1775200676.552125883] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200676.552642090] [action_service]: "actions": ['wait(3)']
[action_service-4] [INFO] [1775200690.343919110] [action_service]: Published message: Robot feedback: Execute wait(3.0) completed
[model_service-3] [INFO] [1775200693.699834863] [model_service]: "json_str": {"response": "Time's up! Now, I'm going to move downward by 3 centimeters—smoot
h and steady, just like a slow descent.", "action": ["arm_move('down', 3)"]}
[model_service-3] [INFO] [1775200693.701079332] [model_service]: "action": ["arm_move('down', 3)"], "response": Time's up! Now, I'm going to move downward b
y 3 centimeters—smooth and steady, just like a slow descent.
[action_service-4] [INFO] [1775200693.707094030] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200693.707823404] [action_service]: "actions": ["arm_move('down', 3)"]
[action_service-4] [INFO] [1775200702.888654190] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.893407678] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.897793997] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.901859240] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.906301244] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.910175598] [action_service]: cur_joints[i-1]: None
[action_service-4] [INFO] [1775200702.910802453] [action_service]: Dir: down
[action_service-4] [INFO] [1775200702.911537698] [action_service]: Dist: 3
[action_service-4] [INFO] [1775200702.912177719] [action_service]: cur_joints: [0, 0, 0, 0, 0, 0]
[kinemarics_dofbot-1] [WARN] [1775200702.918310881] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[kinemarics_dofbot-1] [INFO] [1775200702.918673679] [rclcpp]: fk sent***
[action_service-4] [INFO] [1775200704.915148529] [action_service]: CurEndPos: [0.0020449992274857477, -0.06338999792660725, 0.040402443637541426, 1.57080366
10372729, 3.650393812038131e-08, -1.5707926657577578]
[action_service-4] [INFO] [1775200704.915820681] [action_service]: request: dofbot_interface.srv.Kinemarics_Request(tar_x=0.0020449992274857477, tar_y=-0.06
338999792660725, tar_z=0.010402443637541427, roll=1.5708036610372729, pitch=3.650393812038131e-08, yaw=-1.538546913711967, cur_joint1=0.0, cur_joint2=0.0, c
ur_joint3=0.0, cur_joint4=0.0, cur_joint5=0.0, cur_joint6=0.0, kin_name='ik')
[kinemarics_dofbot-1] [WARN] [1775200704.919417126] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a r
oot link with an inertia.  As a workaround, you can add an extra dummy link to your URDF.
[action_service-4] [INFO] [1775200706.424566003] [action_service]: joints: [0.0, -86, 155, 291, 90, 30]
[action_service-4] [INFO] [1775200708.427442979] [action_service]: Moving.
[action_service-4] [INFO] [1775200708.431167720] [action_service]: self.combination_mode: False
[action_service-4] [INFO] [1775200708.431694927] [action_service]: self.interrupt_flag: False
[action_service-4] [INFO] [1775200708.432627450] [action_service]: Move done.
[action_service-4] [INFO] [1775200708.433415899] [action_service]: Published message: Robot feedback: Execution arm_move_done() completed
[model_service-3] [INFO] [1775200710.524737907] [model_service]: "json_str": {"response": "Downward movement complete! Now, I'll take another brief pause—th
ree seconds of calm before the next action.", "action": ["wait(3)"]}
[model_service-3] [INFO] [1775200710.525396558] [model_service]: "action": ['wait(3)'], "response": Downward movement complete! Now, I'll take another brief
 pause—three seconds of calm before the next action.
[action_service-4] [INFO] [1775200710.534062270] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200710.534598921] [action_service]: "actions": ['wait(3)']
[action_service-4] [INFO] [1775200722.187367767] [action_service]: Published message: Robot feedback: Execute wait(3.0) completed
[model_service-3] [INFO] [1775200724.764601885] [model_service]: "json_str": {"response": "Alright, time to shake things up a bit—literally! I'm going to sh
ake my head now, just like a robot doing its best 'no' dance.", "action": ["arm_shake()"]}
[model_service-3] [INFO] [1775200724.765178962] [model_service]: "action": ['arm_shake()'], "response": Alright, time to shake things up a bit—literally! I'
m going to shake my head now, just like a robot doing its best 'no' dance.
[action_service-4] [INFO] [1775200724.769843097] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200724.770289488] [action_service]: "actions": ['arm_shake()']
[action_service-4] [INFO] [1775200742.607277555] [action_service]: Published message: Robot feedback: Execute arm_shake() completed
[model_service-3] [INFO] [1775200746.450718174] [model_service]: "json_str": {"response": "Head shake complete! Now, I'm going to adjust my joint 6 to 150 d
egrees—precision at its finest!", "action": ["adjust_joint(6, 150)"]}
[model_service-3] [INFO] [1775200746.451214399] [model_service]: "action": ['adjust_joint(6, 150)'], "response": Head shake complete! Now, I'm going to adju
st my joint 6 to 150 degrees—precision at its finest!
[action_service-4] [INFO] [1775200746.455631885] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200746.456283295] [action_service]: "actions": ['adjust_joint(6, 150)']
[action_service-4] [INFO] [1775200758.768324659] [action_service]: Published message: Robot feedback: Execution adjust_joint_done() completed
[model_service-3] [INFO] [1775200761.954300553] [model_service]: "json_str": {"response": "All done! I've completed every step—up, pause, down, pause, shake
, pause, and finally adjusted joint 6 to 150 degrees. Mission accomplished!", "action": ["finishtask()"]}
[model_service-3] [INFO] [1775200761.955190428] [model_service]: "action": ['finishtask()'], "response": All done! I've completed every step—up, pause, down
, pause, shake, pause, and finally adjusted joint 6 to 150 degrees. Mission accomplished!
[action_service-4] [INFO] [1775200761.960818642] [action_service]: "len(actions)": 1
[action_service-4] [INFO] [1775200761.962118205] [action_service]: "actions": ['finishtask()']
```

Core Code Analysis

Please refer to the **Core Code Analysis** content in the tutorial 【AI Model-Text Version】-【3. Text Large Model + robotic arm control】. The voice version and text version have the same action functions; only the input method for task commands is different.